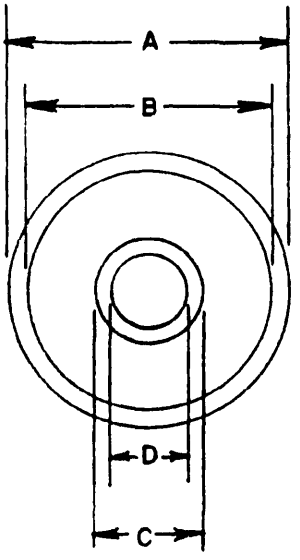
MIL-L-3890/2A
23 January 1976
SUPERSEDING
MIL-L-3890/2
13 June 1962

MILITARY SPECIFICATION

LINES, RADIO FREQUENCY TRANSMISSION (COAXIAL, AIR DIELECTRIC) 75 OHM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring transmission lines, described herein shall consist of this document and the latest issue of Specification MIL-L-3890.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. Transmission line.

FSC 5985

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TABLE I. Dash numbers and dimensions.

Dash number	Line size	Outer conductor					Inner conductor				
		Mat'l.	Dia		Wall thickness 1/		Mat'l.	Dia		Wall thickness 1/	
			A	B	Nom	Max deviation from average 2/		C	D	Nom	Max deviation from average 2/
01 ___	6.125 (155.58)	AL	6.125 (155.58) ±0.008 (.20)	5.981 (151.92) ±0.008 (.20)	0.072 (1.83)	0.012 (.30)	CU	1.711 (43.46) ±0.004 (0.10)	1.661 (42.19) ±0.004 (0.10)	0.025 (0.64)	0.006 (0.15)
02 ___		CU									
03 ___	3.125 (79.38)	AL	3.125 (79.38) ±0.005 (.13)	3.027 (76.89) ±0.005 (.13)	0.049 (1.24)	0.0075 (.19)		0.8690 (22.07) ±0.0025 (0.06)	0.7990 (20.29) ±0.0025 (0.06)	0.035 (0.89)	0.005 (0.13)
04 ___		CU									
05 ___	1.625 (41.28)	AL	1.625 (41.28) ±0.003 (.08)	1.527 (38.79) ±0.003 (.08)	0.049 (1.24)	0.005 (.13)		0.438 (11.13) ±0.002 (0.05)	0.378 (9.60) ±0.002 (0.05)	0.030 (0.76)	0.003 (0.08)
06 ___		CU									
07 ___	0.875 (22.23)	AL	0.8750 (22.23) ±0.0025 (.06)	0.7850 (19.94) ±0.0025 (.06)	0.045 (1.14)	0.005 (.13)		0.226 (5.74) ±0.002 (0.05)	0.176 (4.47) ±0.002 (0.05)	0.025 (0.64)	0.003 (0.08)
08 ___		CU					CU				

1/ Average wall thickness is one-half the difference between the corresponding inside and outside diameters at any cross section perpendicular to the axis, and measured arbitrarily at any point on the periphery.

2/ Maximum deviation of the wall thickness from the average wall thickness (eccentricity) at any cross section is the difference between the average wall thickness and either the maximum or minimum wall thickness, whichever is greater.

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TABLE II. Electrical characteristics.

Dash number		Line size	Operating frequency range	Characteristic impedance	Dielectric withstanding voltage	Attenuation (dB/100 ft) 2 MHz	10 MHz	30 MHz
				Ohms	kV (rms)			
01---	AL	6.125	.5 MHz - 1.0 GHz	75 $^{+.23}_{-.69}$	40.0	.008	.012	.013
02---	CU					.007	.011	.012
03---	AL	3.125	.5 MHz - 1.8 GHz	75 $^{+.12}_{-.41}$	20.0	.014	.032	.056
04---	CU					.013	.030	.054
05---	AL	1.625	.5 MHz - 3.0 GHz	75 $^{+.29}_{-.49}$	10.0	.056	.127	.221
06---	CU					.054	.120	.210
07---	AL	.875	.5 MHz - 3.7 GHz	75 $^{+.41}_{-1.04}$	5.0	.024	.056	.100
08---	CU					.023	.054	.096

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REQUIREMENTS:

Design, construction, and physical dimensions: See figure 1 and table I.
Operating frequency range: See table II.

Attenuation: See table II.

Voltage standing wave ratio (VSWR): 1.10:1.

Characteristic impedance: See table II.

Velocity of propagation: 99.7 percent, minimum.

Dielectric withstanding voltage: See table II.

Pressurization:
Test: 30 lb_f/in^2 $^{+2}_{-0}$ guage

Operating: 20 lb_f/in^2 ±2 guage.

Part number: M3890/2- (and dash number from table I.)

Example: M 3890/1 -01 010

Military designator
Specification sheet number
Dash number
length in inches

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MU, MI, SG, ME
Navy - OS
Air Force - 11, 17
DSA - ES

User activities:
Army - AT, WC
Navy - MC, SH, AS
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5985-0815-2)

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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

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